

Exam Date & Time: 06-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Semiconductor Device Theory [ECE-4064]

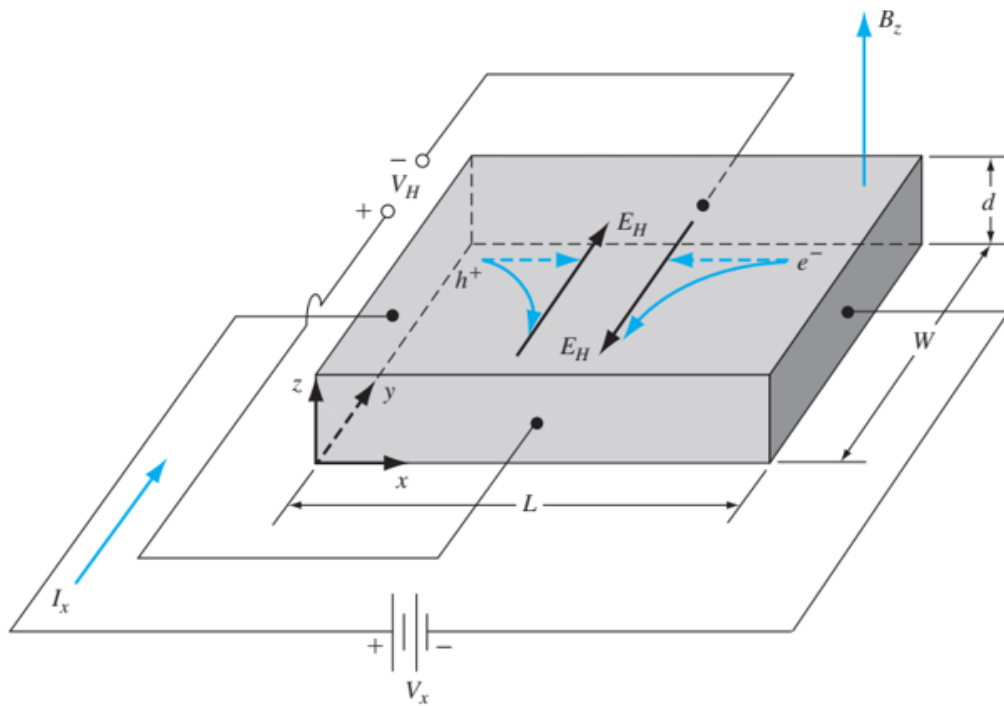
Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1)
 - (i) Determine the probability that an energy level is occupied by an electron if the state is above Fermi level by (a) kT , (b) $5kT$ and (c) $10kT$.
A) (4)
 - (ii) Assume that the Fermi energy level for a particular material is 6.25 eV and that the electrons in this material follow the Fermi–Dirac distribution function. Calculate the temperature at which there is a 1 percent probability that a state 0.30 eV below the Fermi energy level will not contain an electron. (4)
- B) A semiconductor has $N_c = 10^{19} \text{ cm}^{-3}$, $N_v = 5 \times 10^{18} \text{ cm}^{-3}$, and $E_g = 2 \text{ eV}$. If it is doped with 10^{17} donors (fully ionized), calculate the intrinsic carrier concentration and hole concentration at 627°C. Sketch the simplified band diagram, showing the position of E_F . (3)
- C) Consider a GaAs semiconductor at $T = 300 \text{ K}$. Calculate the thermal equilibrium electron and hole concentrations for (i) $N_d = 2 \times 10^{15} \text{ cm}^{-3}$, $N_a = 0$, and (ii) $N_a = 10^{16} \text{ cm}^{-3}$, $N_d = 7 \times 10^{15} \text{ cm}^{-3}$. (3)
- 2)
 - Consider a gallium arsenide sample at $T = 300 \text{ K}$. A Hall effect device has been fabricated with the following geometry: $d = 0.01 \text{ cm}$, $W = 0.05 \text{ cm}$, and $L = 0.5 \text{ cm}$. The electrical parameters are: $I_x = 2.5 \text{ mA}$, $V_x = 2.2 \text{ V}$, and $B_z = 2.5 \times 10^{-2} \text{ tesla}$. The Hall voltage is $V_H = -4.5 \text{ mV}$. Find: (a) the conductivity type, (b) the majority carrier concentration, and (c) the mobility. (4)



- B) A silicon pn junction at $T = 300$ K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16} \text{ cm}^{-3}$ and $N_a = 5 \times 10^{15} \text{ cm}^{-3}$. Determine (a) x_n , (b) x_p , (c) W , and (d) E_{max} . (3)
- C) A silicon pn junction at $T = 300$ K has doping concentrations of $N_d = 3 \times 10^{16} \text{ cm}^{-3}$ and $N_a = 8 \times 10^{15} \text{ cm}^{-3}$ and has a cross-sectional area of $A = 5 \times 10^{-5} \text{ cm}^2$. Determine the junction capacitance at (a) $V_R = 2$ V and (b) $V_R = 5$ V. (3)
- 3) A Schottky diode is formed by depositing gold on n-type GaAs that is doped at a concentration of $N_d = 5 \times 10^{15} \text{ cm}^{-3}$. For $T = 300$ K, determine the theoretical values of (a) ϕ_{B0} , (b) $\phi_{n\phi}$, (c) V_{bi} , and (d) x_n and E_{max} at $V_R = 5$ V. (4)
- A) A Schottky barrier diode formed on n-type silicon has a doping concentration of $N_d = 5 \times 10^{15} \text{ cm}^{-3}$ and a barrier height of $\phi_{B0} = 0.65$ V. Determine the built-in potential barrier V_{bi} . If the doping concentration changes to $N_d = 10^{16} \text{ cm}^{-3}$, determine the values of ϕ_{B0} and V_{bi} . Do these values increase, decrease, or remain the same? (3)
- B) Distinguish between the Schottky diode and the p-n junction diode with an appropriate band diagram. (3)
- 4) (4)
- A) (4)

Calculate the internal pinch-off voltage and pinch-off voltage of n-channel JFET. Assume that the p-n junction of a uniformly doped silicon n-channel JFET at $T=300$ K has doping concentrations of $N_a=10^{18} \text{ cm}^{-3}$ and $N_d=10^{16} \text{ cm}^{-3}$. Assume that the metallurgical channel thickness, a , is $0.75 \mu\text{m}$.

- B) Consider an n-channel GaAs MESFET at $T=300$ K with a gold Schottky barrier contact. Assume the barrier height is $\phi_{Bn} = 0.89$ V. The n-channel doping is $N_d=2 \times 10^{15} \text{ cm}^{-3}$. Design the channel thickness such that $V_T=0.25$ V. (3)
- C) Calculate C_{ox} , C'_{min} , and C'_{FB} for a MOS capacitor. Consider a p-type silicon substrate at $T = 300$ K doped to $N_a = 10^{16} \text{ cm}^{-3}$. The oxide is silicon dioxide with a thickness of $t_{ox} = 18$ nm, and the gate is aluminium. (3)
- 5) Determine the metal–semiconductor work function difference and the threshold voltage for a silicon MOS device at $T = 300$ K for the following parameters: p+ polysilicon gate, $N_a = 2 \times 10^{16} \text{ cm}^{-3}$, $t_{ox} = 8$ nm, and $Q'_{ss} = 2 \times 10^{10} \text{ cm}^{-2}$. (4)
- A) An n-channel silicon MOSFET has the following parameters: $W = 6 \mu\text{m}$, $L = 1.5 \mu\text{m}$, and $t_{ox} = 8$ nm. When the transistor is biased in the saturation region, the drain current is $I_{D(sat)} = 0.132$ mA at $V_{GS} = 1.0$ V and $I_{D(sat)} = 0.295$ mA at $V_{GS} = 1.25$ V. Determine the electron mobility and the threshold voltage. (3)
- B) Silicon MOSFET has the following parameters: $N_a = 10^{16} \text{ cm}^{-3}$ and $t_{ox} = 12$ nm. Calculate (a) the body-effect coefficient and (b) the change in threshold voltage for $V_{SB} = 1$ V. (3)

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